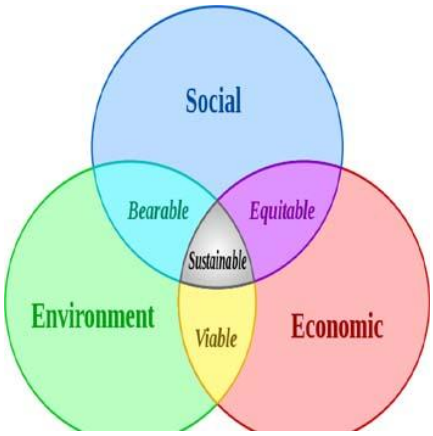
Environmental Science: Handout
Gono Bishwabidyalay

1. Define environment.

In simple words, environment is defined as the natural world in which people, animals and plants live. The term Environment is derived from French terms '*virer* (to turn) or 'in/viron' meaning to encircle; and denotes the interaction between natural surroundings and organisms including human beings. Thus, environment is the surroundings in which living and non-living things live, interact, grow and perish. The conditions present in the surroundings govern the life of creatures therein.



This figure is a **Venn diagram** illustrating the **three pillars of sustainability**—**Social**, **Economic**, and **Environmental**—as taught in Environmental Science. It is a foundational concept for understanding sustainable development and human-environment interaction. This represents **sustainable development**, defined by the Brundtland Commission (1987) as: "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

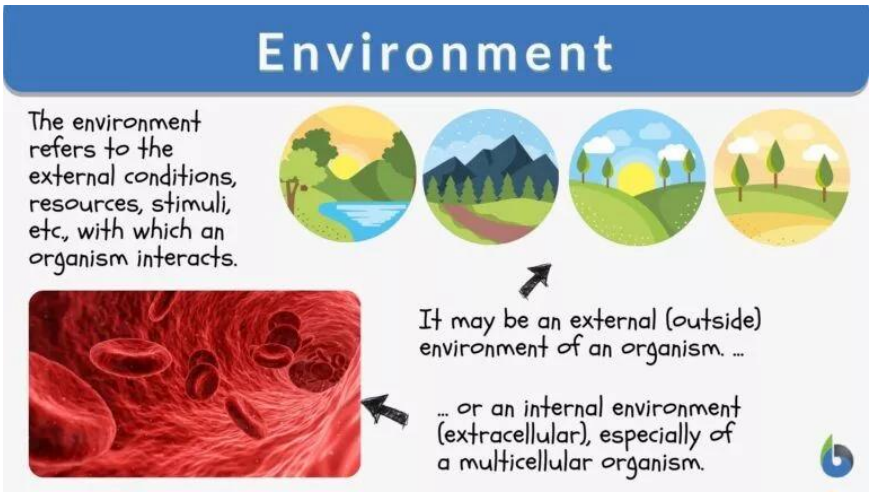
Intersection	Description
Social + Environmental = Bearable	Conditions that support both social well-being and environmental protection. Example: Green public parks that improve both community life and biodiversity.
Social + Economic = Equitable	Fair distribution of wealth and opportunities. Example: Jobs that are both well-paid and socially responsible.
Economic + Environmental = Viable	Projects that are both economically sound and environmentally friendly. Example: Renewable energy businesses.

Some important definitions of environment are as under:

(1) According to Boring (1984), 'A person's environment consists of the sum total of the stimulation which he receives from his conception until his death.' Indicating that environment comprises various types of forces such as physical, intellectual, mental, economical, political, cultural, social, moral and emotional.

(2) Douglas and Holland (1947) defined that 'The term environment is used to describe, in aggregate, all the external forces, influences and conditions, which affect the life, nature, behaviour and the growth, development and maturity of living organisms'.

(3) Woodworth and Marquis (1948) said, 'Environment covers all the outside factors that acted on the individual since he began life."



Type	Description	Example
External Environment	Everything **outside** of the body or organism.	Mountains, forests, rivers, air
Internal Environment	The **inside** space of a living body where cells live and work together.	Blood, body fluids (e.g., plasma)

2. Definition of Environmental Science

Environmental science is an interdisciplinary subject. It draws from a whole gamut of subjects like geography, geology, biology, chemistry, physics, botany, zoology, genetics, sociology, rural development, urban planning, policy development, politics, cultural studies, economics, ethics, law, education, health, communication and philosophy. It is a science that will address the biggest challenges encountering

the planet in this millennium. It studies the complex interaction of human with nature and provides guidelines for the way forward.



3. Basic Components of Environment

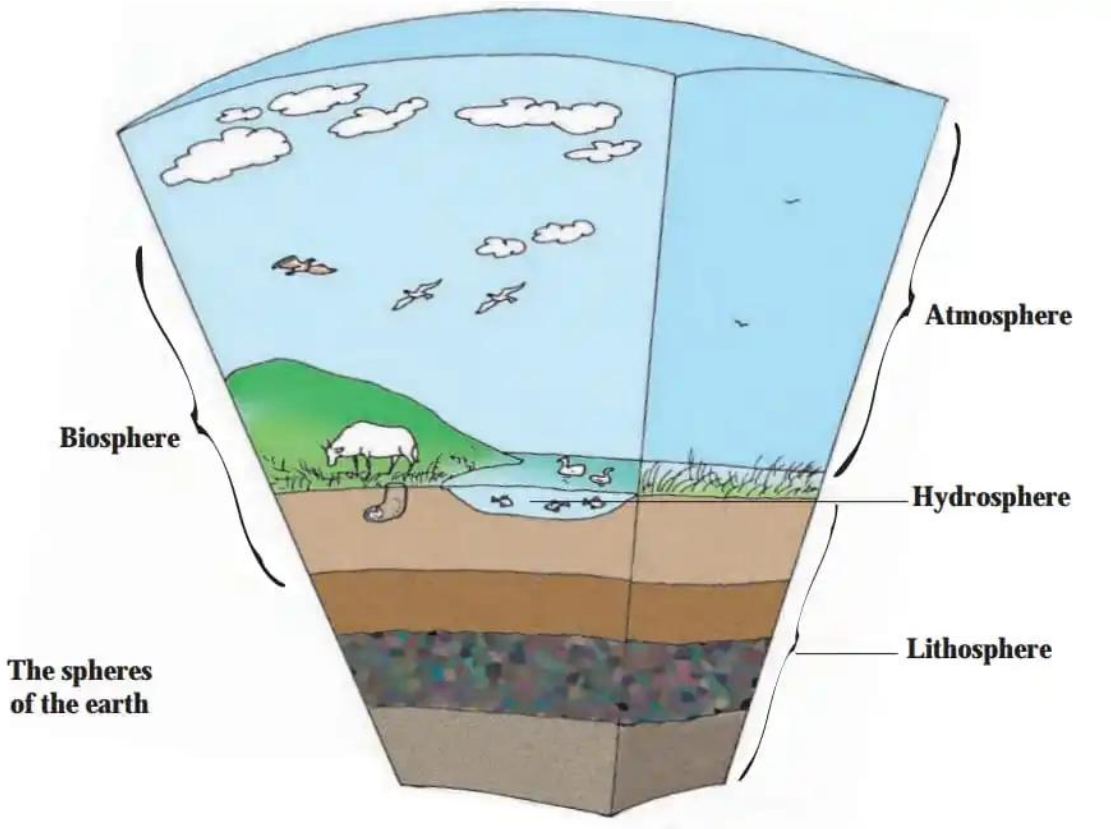
The four major components of environment include lithosphere, hydrosphere, atmosphere and biosphere, corresponding to rocks, water, air and life respectively.

a. **Lithosphere** is the outermost layer of earth called crust, which is made of different minerals. Its depth can reach up to 100 kilometers and is found on both land (terrestrial crust) and oceans (oceanic crust). The main component of lithosphere is earth's tectonic plates.

b. **Hydrosphere** comprises of all forms of water bodies on earth including oceans, seas, rivers, lakes, ponds, streams etc. It covers 70% of earth's surface. 97.5% of water found on Earth is in the oceans in the form of salt water. Only 2.5 % of water on Earth is freshwater. Out of this, 30.8% is available as groundwater and 68.9% is in frozen forms as in glaciers. Amount of 0.3% is available in rivers, reservoirs and lakes and is easily accessible to man.

c. **Atmosphere** is gaseous layer enveloping the Earth. The atmosphere with oxygen in abundance is unique to Earth and sustains life. It mainly comprises 78.08% nitrogen, 20.95% oxygen, 0.93% argon, 0.038% carbon dioxide, and traces of hydrogen, helium, and noble gases. The amount of water vapour present is variable.

d. **Biosphere** refers to all the regions on Earth where life exists. The ecosystems that support life could be in soil, air, water or land. The term Biosphere was coined by Geologist Edward Suess who used this term for place on Earth where life can be found. Biosphere refers to the sum total of all living matter, the biomass or biota. It extends from the polar ice caps to the equator, with each region harboring some life form suitable to the conditions there.



4. Outline of environmental studies

- Aquatic and environmental engineering
- Biomimetics (Biomimicry)
- Climatology
- Conservation biology
- Conservation movement
- Ecocriticism
- Ecological economics
- Ecological engineering
- Ecological genetics
- Ecological humanities
- Ecological literacy
- Ecological psychology
- Energy and environment
- Energy conservation
- Environmental archaeology
- Environmental chemistry
- Environmental degradation
- Environmental design
- Environmental economics
- Environmental effects on physiology
- Environmental engineering
- Environmental ethics
- Environmental finance

- Environmental geography
- Environmental geology
- Environmental history
- Environmental impact assessment
- Environmentalism
- Environmental issues
- Environmental justice
- Environmental law
- Environmental management
- Environmental monitoring
- Environmental movement
- Environmental organization
- Environmental philosophy
- Environmental psychology
- Environmental policy
- Environmental protection
- Environmental racism
- Environmental remediation
- Environmental restoration
- Environmental science
- Environmental sociology
- Environmental soil science
- Environmental studies
- Environmental technology
- Environmental toxicology
- Geomatics
- Green chemistry
- Green computing
- Green economy
- Green engineering
- Landscape architecture
- Natural resource management
- Fishing
- Hunting
- Hydrology
- Oceanography
- Pollutant
- Pollution
- Renewable energy
- Renewable resource
- Restoration ecology
- Sustainability science
- Sustainability studies
- Toxicology
- Traditional environmental knowledge
- Waste management
- Waste minimisation
- Whaling
- Wildlife conservation
- Wildlife management
- Wildlife observation

5. Scope of environmental studies

What does Environmental Science cover, that is known as scope. The scope of environmental studies is very wide and it deals with many areas like:

i) Conservation of natural resources,

ii) Ecological aspects,

iii) Pollution of the surrounding natural resources,

iv) Controlling the pollution,

v) Social issues connected to it, and

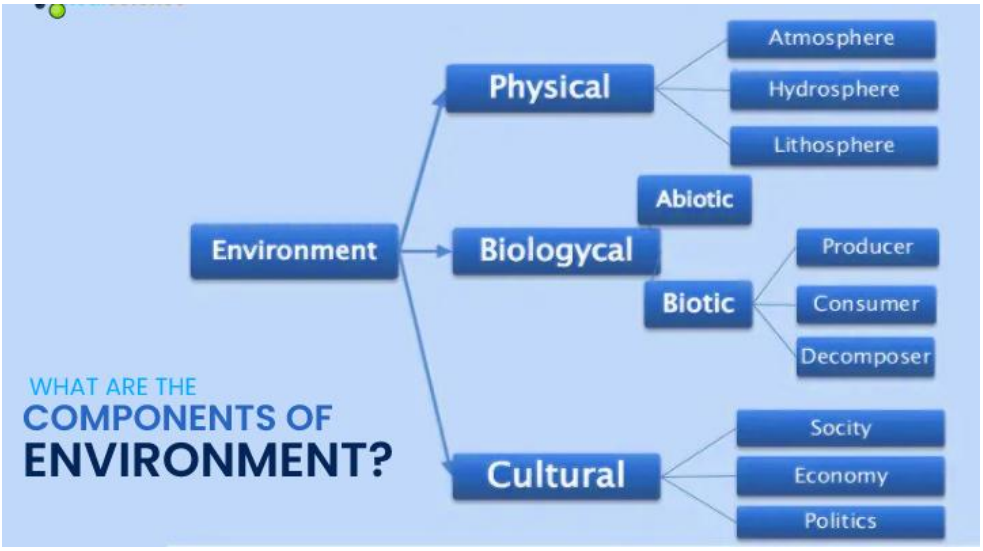
vi) Impacts of human population on the environment.



6. Elements vs Components of Environment

Environment is constituted by the interacting systems of physical, biological and cultural elements inter-related in various ways, individually as well as collectively. These elements are:

(1) Physical elements: Physical elements are space, landforms, water bodies, climate, soils, rocks and minerals. They determine the variable character of the human habitat, its opportunities as well as limitations.
(2) Biological elements: Biological elements such as plants, animals, microorganisms and men constitute the biosphere.
(3) Cultural elements: Cultural elements such as economical, social and political elements are essentially man- made features, which make the cultural background.



7. Importance of environmental studies

The environment studies make us aware about the importance of protection and conservation of our mother earth and about the destruction due to the release of pollution into the environment. The increase in human and animal population, industries and other issues make the survival cumbersome. A great number of environment issues have grown in size and make the system more complex day by day, threatening the survival of mankind on earth. Environment studies have become significant for the following reasons:

1. Environment Issues are being of Global:

It has been well recognised that environment issues like global warming and ozone depletion, acid rain, marine pollution and biodiversity are not merely national issues but are global issues and hence require international efforts and cooperation to solve them.

2. Development and Environment:

Development leads to Urbanization, Industrial Growth, Telecommunication and Transportation Systems, Hi-tech Agriculture and Housing etc. However, it has become phased out in the developed world. The North intentionally moves their dirty factories to South to cleanse their own environment. When the West developed, it did so perhaps in ignorance of the environmental impact of its activities. Development of the rich countries of the world has undesirable effects on the environment of the entire world.

3. Explosive Increase in Pollution

World census reflects that one in every seven persons in this planet lives in India. Evidently with 16 per cent of the world's population and only 2.4 per cent of its land area, there is a heavy pressure on the natural resources including land.

Agricultural experts have recognized soil health problems like deficiency of micronutrients and organic matter, soil salinity and damage of soil structure.

4. Need for an Alternative Solution

It is essential, specially for developing countries to find alternative paths to an alternative goal. We need a goal as under:

- A true goal of development with an environmentally sound and sustainable development.
- A goal common to all citizens of our planet earth.
- A goal distant from the developing world in the manner it is from the over-consuming wasteful societies of the "developed" world.

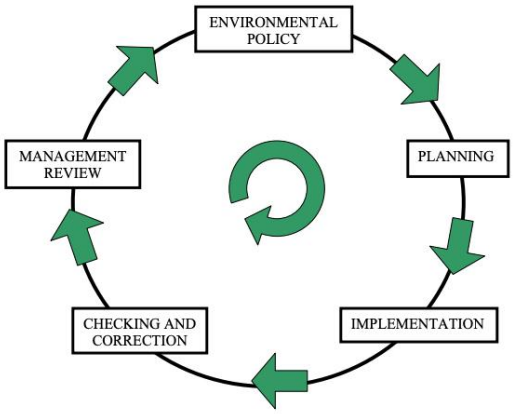
It is utmost important for us to save the humanity from extinction because of our activities constricting the environment and depleting the biosphere, in the name of development.

5. Need for Wise Planning of Development

Our survival and sustenance depend on resources availability. Hence Resources withdraw, processing and use of the products have all to be synchronised with the ecological cycle. In any plan of development our actions should be planned ecologically for the sustenance of the environment and development.

8. Natural resource management

Natural resource management issues are inherently complex and contentious. First, they involve the ecological cycles, hydrological cycles, climate, animals, plants and geography, etc. All these are dynamic and inter-related. A change in one of them may have far reaching and/or long term impacts which may even be irreversible. Second, in addition to the complexity of the natural systems, managers also have to consider various stakeholders and their interests, policies, politics, geographical boundaries and economic implications. It is impossible to fully satisfy all aspects at the same time. Therefore, between the scientific complexity and the diverse stakeholders, natural resource management is typically contentious. The following diagram represents the **Environmental Management System (EMS) Cycle**, a key framework used in **Natural Resource Management (NRM)**. It is a continuous improvement process used by organizations, governments, and institutions to manage environmental performance in a structured and sustainable way.

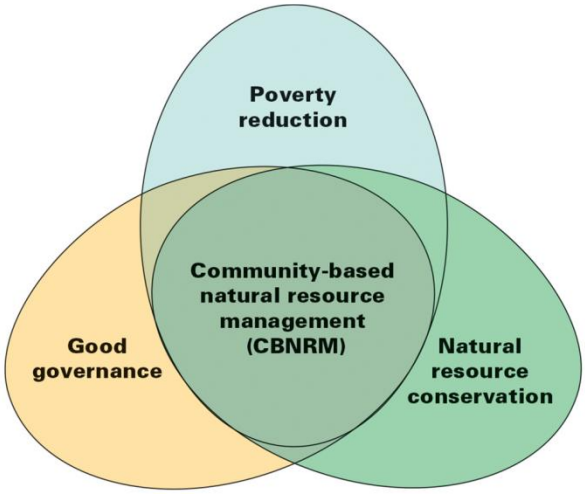


Step	Description
1. **Environmental Policy**	A formal statement of commitment to sustainable natural resource use and environmental protection.
2. **Planning**	Identifying environmental aspects, legal requirements, objectives, and targets. Often involves stakeholder engagement.
3. **Implementation**	Executing the environmental plan (e.g. afforestation, soil conservation, pollution control).
4. **Checking and Correction**	Monitoring, auditing, and identifying any problems or deviations.
5. **Management Review**	Reviewing progress and making improvements or changes for the next cycle.

As environmental challenges grow more complex due to climate change, overpopulation, and industrialization, a diverse and inclusive approach to managing these resources has become essential. To address these challenges effectively, modern NRM practices incorporate a mix of community participation, social inclusion, flexibility, integration, and technology. Among them, the following five approaches play a critical role:

(1) The community-based natural resource management (CBNRM) approach combines conservation objectives with the generation of economic benefits for rural communities. The three key assumptions being that: locals are better placed to conserve natural resources, people will conserve a resource only if benefits exceed the costs of conservation, and people will conserve a resource that is linked

directly to their quality of life. When a local people's quality of life is enhanced, their efforts and commitment to ensure the future well-being of the resource are also enhanced. Regional and community based natural resource management is also based on the principle of subsidiarity.



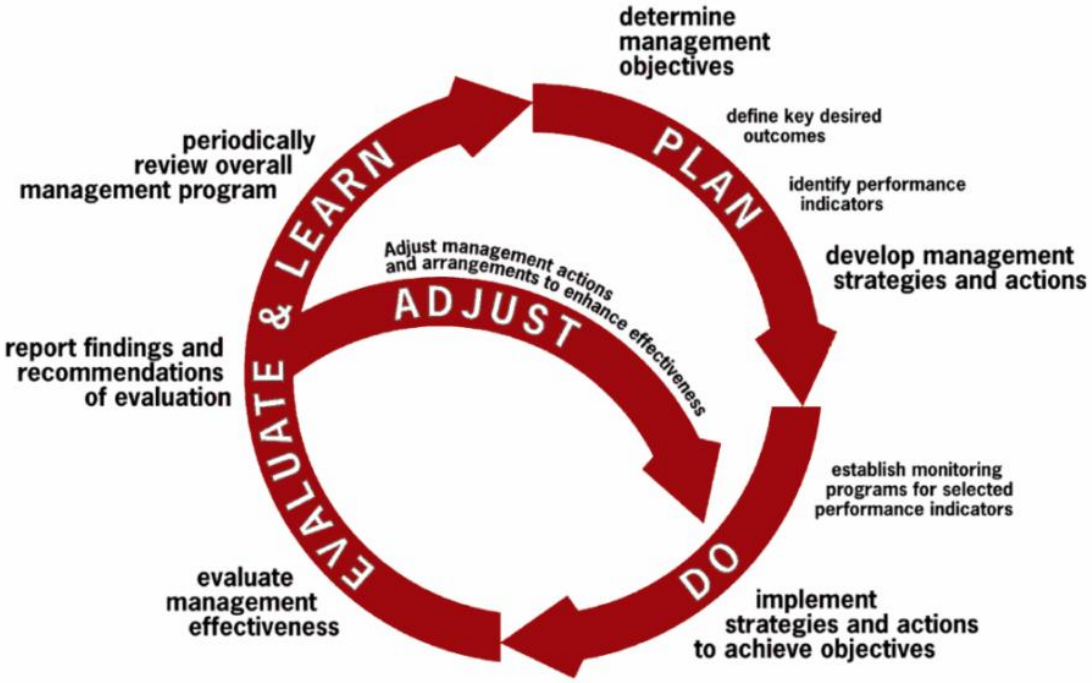
(2) Gender-based natural resource management: Social capital and gender are factors that impact community-based natural resource management (CBNRM), including conservation strategies and collaborations between community members and staff. Through three months of participant observation in a fishing camp in San Evaristo, Mexico, Ben Siegelman learned that the fishermen build trust through jokes and fabrications. He emphasizes social capital as a process because it is built and accumulated through practice of intricate social norms.



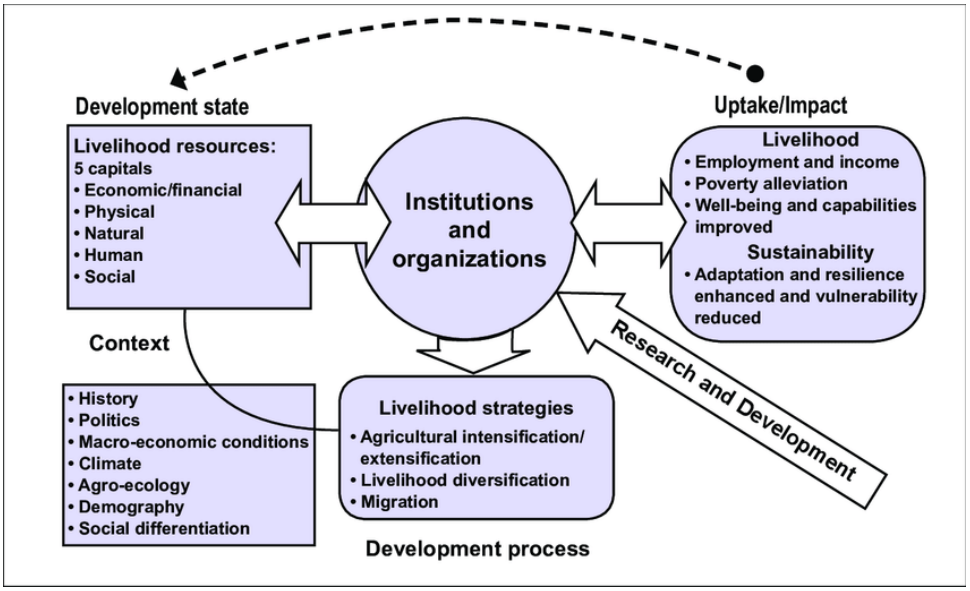
(3) Adaptive Management: This approach includes recognition that adaption occurs through a process of 'plan-do-review-act'. It also recognises seven key components that should be considered for quality natural resource management practice:

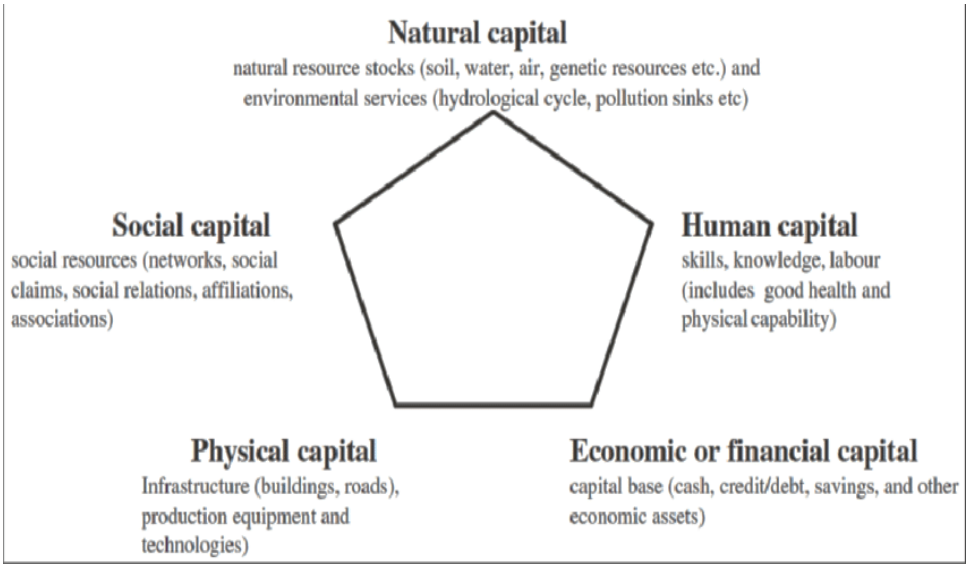
- Determination of scale
- Collection and use of knowledge
- Information management

- Monitoring and evaluation
- Risk management
- Community engagement
- Opportunities for collaboration.

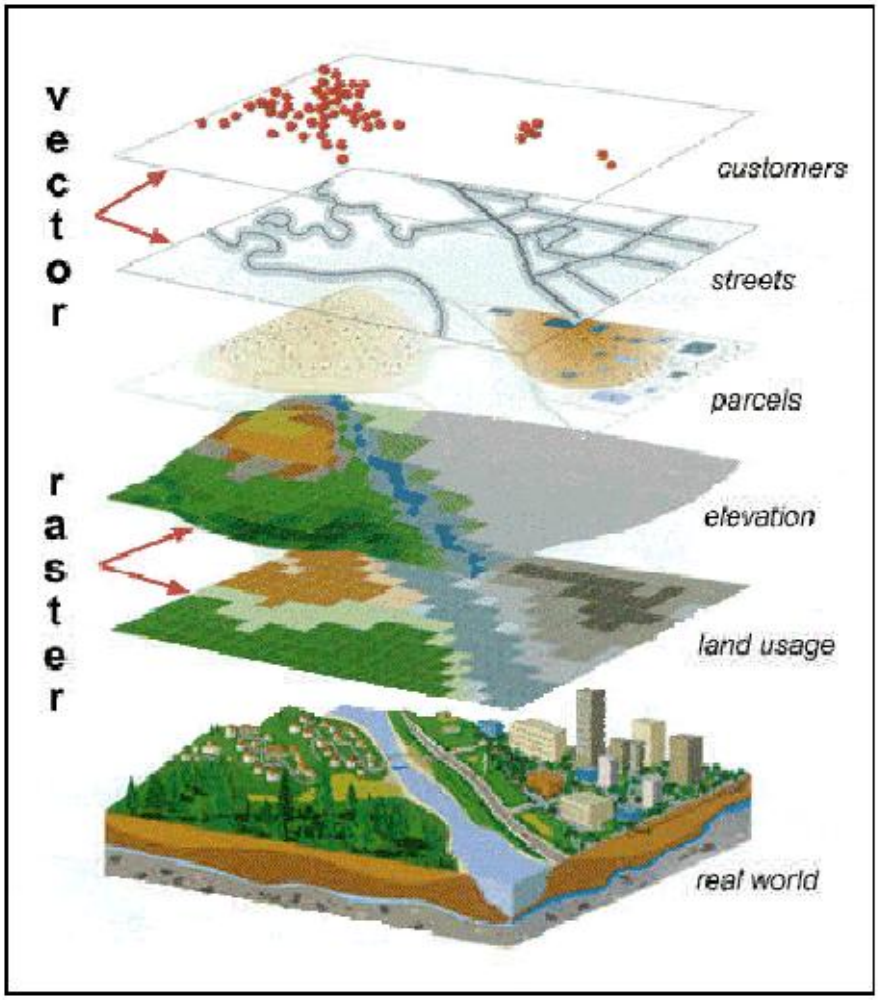


(4) Integrated natural resource management: Integrated natural resource management (INRM) is the process of managing natural resources in a systematic way, which includes multiple aspects of natural resource use (biophysical, socio-political, and economic) meet production goals of producers and other direct users (e.g., food security, profitability, risk aversion) as well as goals of the wider community (e.g., poverty alleviation, welfare of future generations, environmental conservation). It focuses on sustainability and at the same time tries to incorporate all possible stakeholders from the planning level itself, reducing possible future conflicts. The conceptual basis of INRM has evolved in recent years through the convergence of research in diverse areas such as sustainable land use, participatory planning, integrated watershed management, and adaptive management. INRM is being used extensively and been successful in regional and community based natural management.





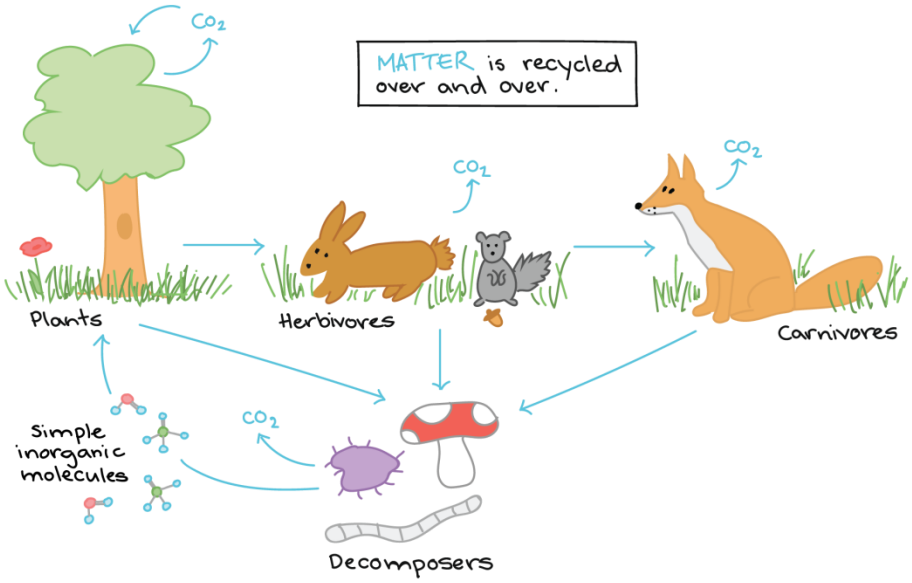
(5) GIS Based Management System: GIS is a powerful analytical tool as it is capable of overlaying datasets to identify links. A bush regeneration scheme can be informed by the overlay of rainfall, cleared land and erosion. It provides data on natural resources such as vegetation, fisheries, soils and water. These are limited by the potential for subjective input and data manipulation.



9. What is Ecosystem?

An ecosystem is a geographic area where plants, animals, and other organisms, as well as weather and landscape, work together to form a bubble of life. Ecosystems contain biotic or living, parts, as well as abiotic factors, or nonliving parts. Biotic factors include plants, animals, and other organisms. Abiotic factors include rocks, temperature, and humidity.

Every factor in an ecosystem depends on every other factor, either directly or indirectly. A change in the temperature of an ecosystem will often affect what plants will grow there, for instance. Animals that depend on plants for food and shelter will have to adapt to the changes, move to another ecosystem, or perish.



Ecosystems can be very large or very small.

MAIN DIFFERENCES

ECOLOGY AND ECOSYSTEM

DEFINITION

Ecology is a branch of biology which deals with the relationships of organisms to one another and to their physical environment	Ecosystem is a community of interacting organisms and their physical environment; an ecosystem is a subpart of ecology

RELATIONSHIP

Ecology includes the study of relationship between living organisms and their environment	Ecosystem is a place like a forest, taiga, grass land, desert, stillwater, river or a stream, coral reefs etc

10. Ecosystem components

The ecosystem has two major kinds of components: (1) Abiotic (non-living) and (2) Biotic (living) components.

(A) Biotic Components of Ecosystem

The living components of an ecosystem are called the biotic components. Some of these factors include plants, animals, as well as fungi and bacteria. These biotic components can be further classified, based on the energy requirement source. Producers, consumers, and decomposers are the three broad categories of biotic components.

Producers are the plants in the ecosystem, which can generate their own energy requirement through photosynthesis, in the presence of sunlight and chlorophyll. All other living beings are dependent on plants for their energy requirement of food as well as oxygen.

Consumers include ***herbivores, carnivores,*** *and* ***omnivores***. The herbivores are the living organisms that feed on plants. Carnivores eat other living organisms. Omnivores are animals that can eat both plant and animal tissue.

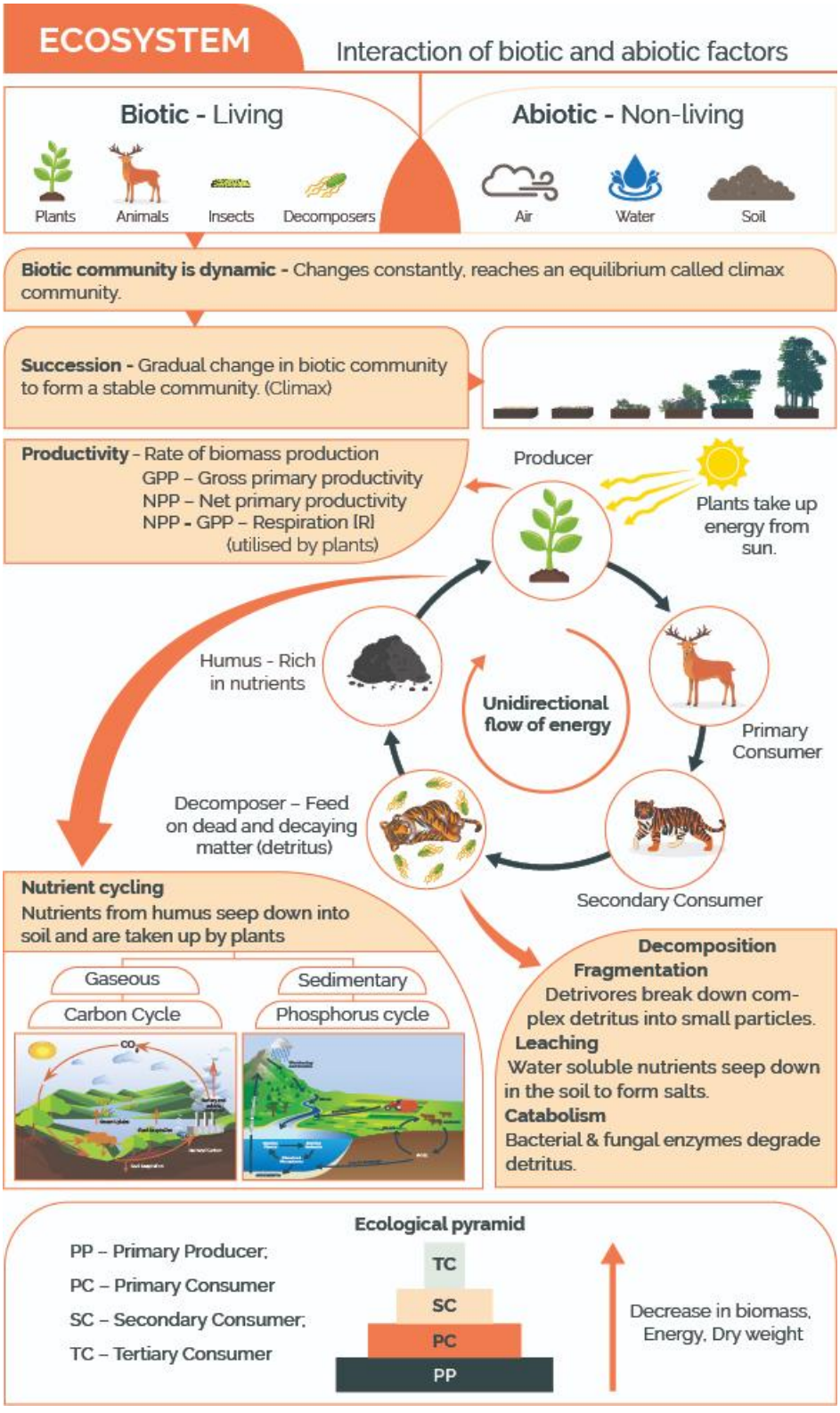
Decomposers are the ***fungi*** *and* ***bacteria,*** which are the ***saprophytes***. They feed on the decaying organic matter and convert this matter into nitrogen and carbon dioxide. The saprophytes play a vital role in recycling the nutrients so that the producers i.e. plants can use them once again.

(B) Abiotic Components of Ecosystem

Abiotic components are the physical and/or the chemical factors that act on the living organisms at any part of their life. These are also called as the ecological factors. The physical and chemical factors are characteristic of the environment. Light, air, soil, and nutrients, etc. form the abiotic components of an ecosystem.

The abiotic factors vary from ecosystem to ecosystem. In an aquatic ecosystem, the abiotic factors may include water pH, sunlight, turbidity, water depth, salinity, available nutrients and dissolved oxygen. Similarly, abiotic factors in terrestrial ecosystems can include soil, soil types, temperature, rain, altitude, wind, nutrients, sunlight etc.

Here, the sun is the energy source. Producers/plants use this energy to synthesize food in the presence of carbon dioxide and chlorophyll. The energy from the sun, through several chemical reactions, turns into chemical energy.

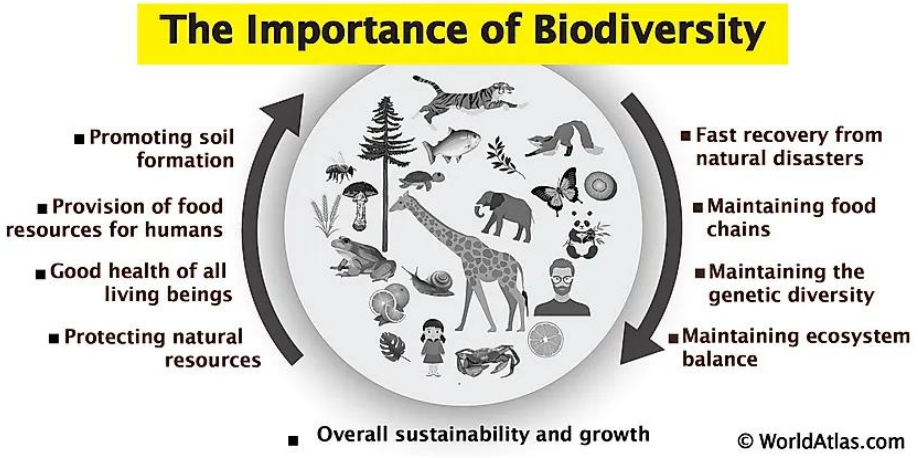


11. Define Biodiversity

Biodiversity or biological diversity is the variety and variability of life on Earth. Biodiversity is a measure of variation at the genetic (genetic variability), species (species diversity), and ecosystem (ecosystem diversity) level.

Biodiversity is not distributed evenly on Earth, it is usually greater in the tropics as a result of the warm climate and high primary productivity in the region near the equator. These tropical forest ecosystems cover less than 10% of earth's surface and contain about 90% of the world's species. Marine biodiversity is usually higher along coasts in the Western Pacific, where sea surface temperature is highest, and in the mid-latitudinal band in all oceans. There are latitudinal gradients in species diversity. Biodiversity generally tends to cluster in hotspots, and has been increasing through time, but will be likely to slow in the future as a primary result of deforestation. It encompasses the evolutionary, ecological, and cultural processes that sustain life.

Rapid environmental changes typically cause mass extinctions. More than 99.9% of all species that ever lived on Earth, amounting to over five billion species, are estimated to be extinct. Estimates on the number of Earth's current species range from 10 million to 14 million, of which about 1.2 million have been documented and over 86% have not yet been described.

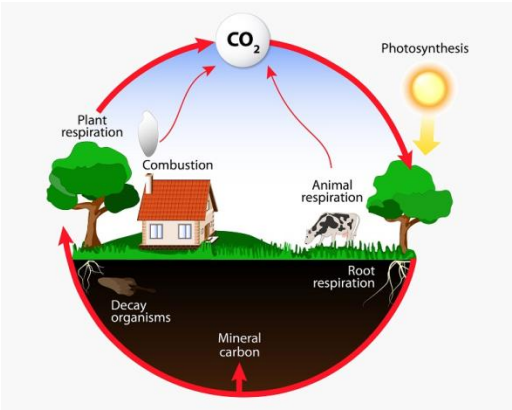


12. Cycles of Matter and Energy Transfer in Ecosystems

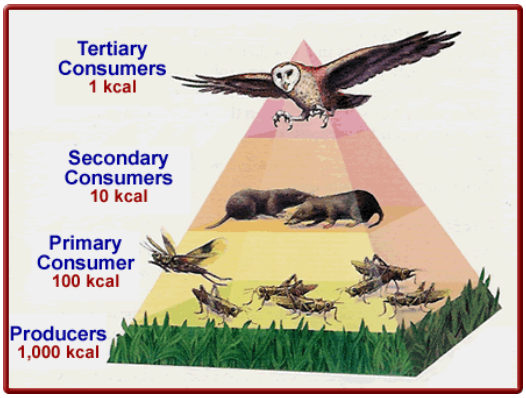
The cycling of matter and the flow of energy within ecosystems occur through interactions among different organisms and between organisms and the physical environment. All living systems need matter and energy. Matter fuels the energy-releasing chemical reactions that provide energy for life functions and provides the material for growth and repair of tissue. Energy from light is needed for plants because the chemical reaction that produces plant matter from air and water requires an energy input to occur. Animals acquire matter from food, that is, from plants or other animals. The chemical elements that make up the molecules of organisms pass through food webs and the environment and are combined and recombined in different ways. At each level in a food web, some matter provides

energy for life functions, some is stored in newly made structures, and much is discarded to the surrounding environment. Only a small fraction of the matter consumed at one level is captured by the next level up. As matter cycles and energy flows through living systems and between living systems and the physical environment, matter and energy are conserved in each change.

The carbon cycle provides an example of matter cycling and energy flow in ecosystems. Photosynthesis, digestion of plant matter, respiration, and decomposition are important components of the carbon cycle, in which carbon is exchanged between the biosphere, atmosphere, oceans, and geosphere through chemical, physical, geological, and biological processes.



Carbon Cycle



Level of consumption

13. Renewable and Non-Renewable Energy

There are two major categories of energy: renewable and non-renewable.

Non-renewable energy resources are available in limited supplies, usually because they take a long time to replenish. The advantage of these non-renewable resources is that power plants that use them are able to produce more power on demand. The non-renewable energy resources are:

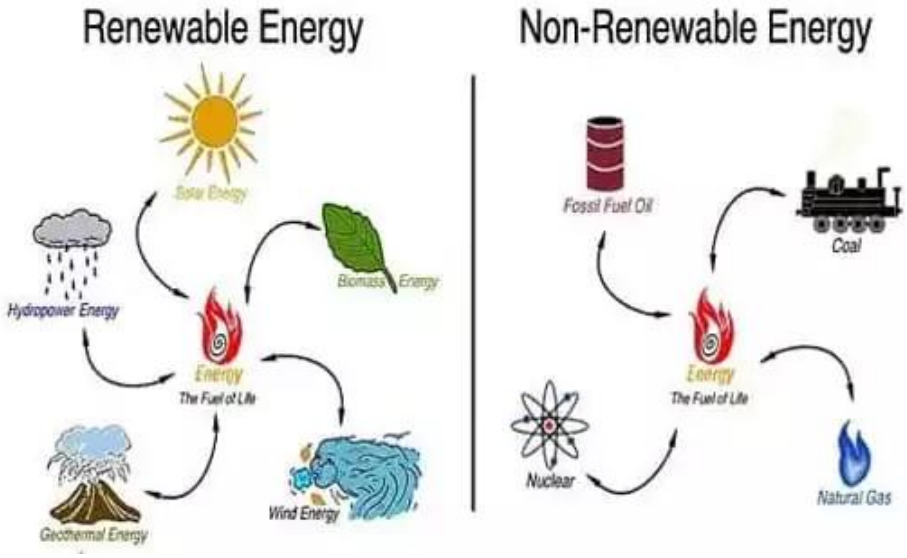
- Coal
- Nuclear
- Oil
- Natural gas

Renewable resources, on the other hand, replenish themselves. The five major renewable energy resources are:

- Solar
- Wind
- Water, also called hydro
- Biomass, or organic material from plants and animals
- Geothermal, which is naturally occurring heat from the earth

While renewable energy resources have the advantage of unlimited supply over the long haul, they are limited in their availability at any given moment.

For example, the sun rises each day, but its ability to generate power is limited when its cloudy. Another disadvantage is that power plant operators can't crank up renewable energy production when people are consuming more power, such as on a hot day when many people are running air conditioners at the same time.

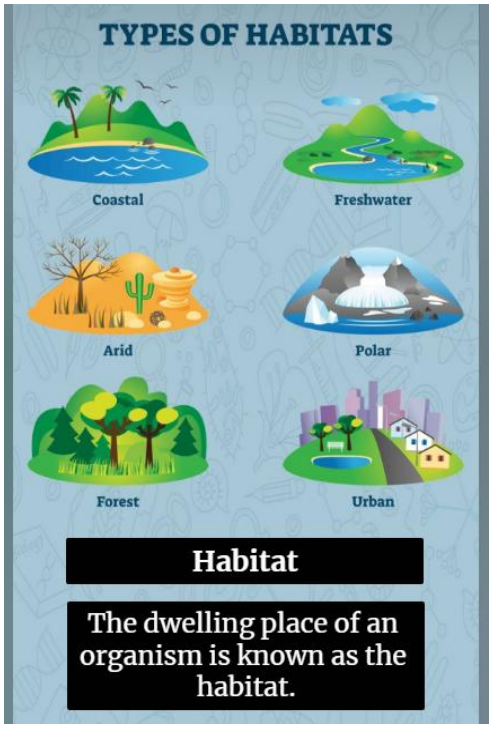


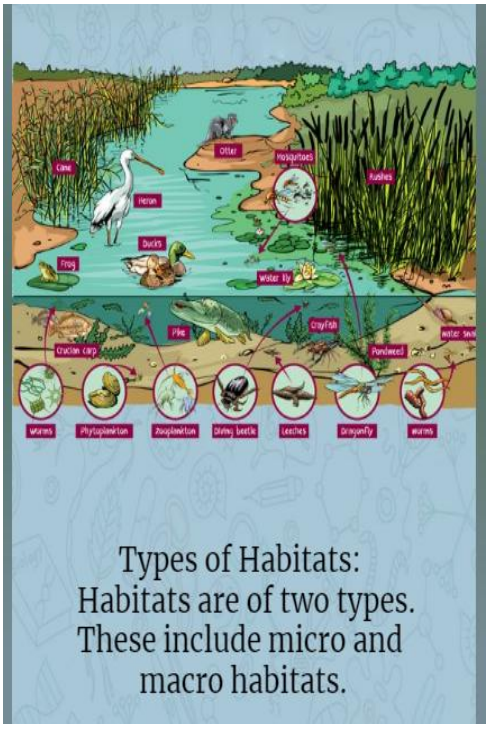
14. Micro Environment or micro-habitat and macro Environment

Habitat is the environment in which a particular organism may live, breed and eventually die. The organism may be either Animal, Plant or Microscopic in nature, while the environment or Physical surroundings provides for its sustainable growth. Organisms may have even multiple habitats based on their physiology and memory. Its commonly practice for Sockeye Salmon to spawn in rivers hundreds of miles inland which is considered their juvenile habitat. However once they reach juvenile maturity, they will venture back out to sea where they will grow to adult maturity and head back to the rivers to spawn and die.

There are 2 main types of Habitats to consider.

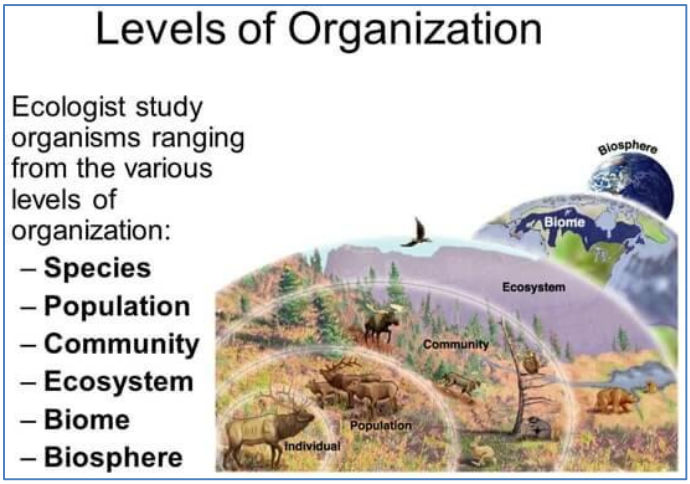
Macro habitat (Is the larger habitat surrounding largely mobile organisms. The macrohabitat may cantain the nescessary elements for multiple organisms to survive and reproduce within its environment). In the image below, the Manatee occupies a macrohabitat in which several other organisms are also dependants upon.

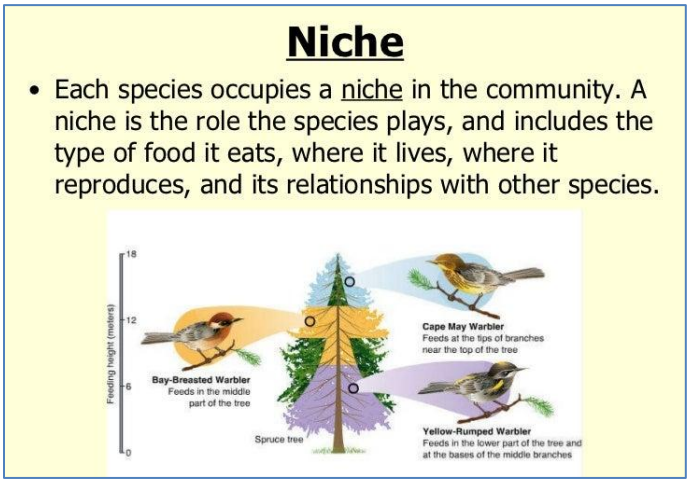




Micro habitat (Is the immediate habitat surrounding small organisms. This microhabitat may not contains all the nutrients needed to support the confined organisms, however owing to the physiological nature of the microhabitat, these organisms will continuously remain in this confined environment) - The following example shows the (A) microhabitat of stringy seaweed below the intertidal zones.

Niche is the term used to define the role of the organism within its habitat. By definition the niche role is played by no more than 1 individual species within its ecological environment. It could also refer to the position or status of an organism within a community and ecosystem resulting from its structural adaptation and behavior. It may also be referred to as an Ecological Niche.

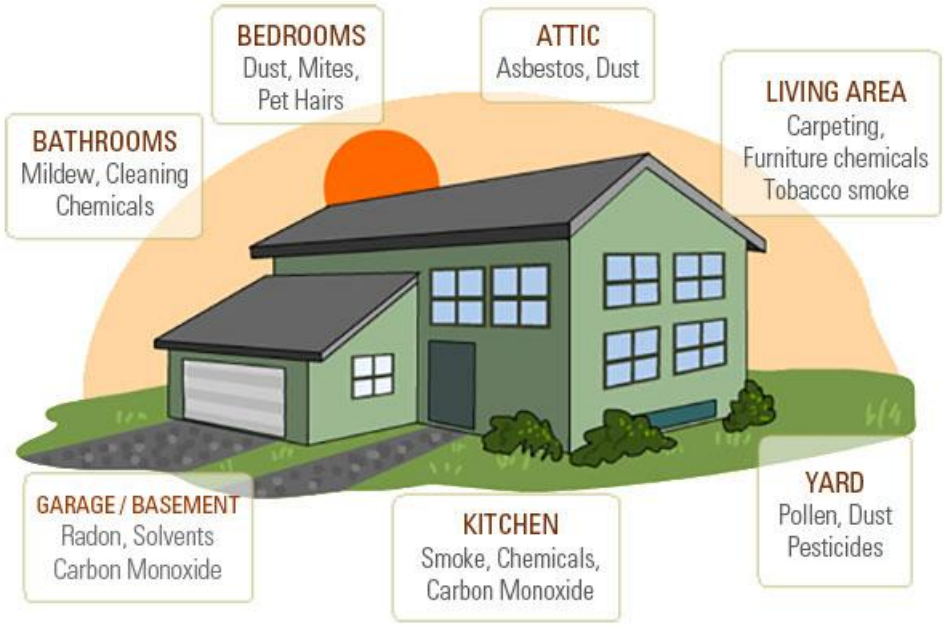




	Ecosystem	Habitat	Microhabitat	Niche
Spatial scale	Large	Small	Much smaller	Cannot be defined in spatial scale
Focal species	Community-centric	Species-centric	Species-centric	Species and individual/sub-group-centric
Foundation of the concept	Empirical	Empirical	Empirical	Theoretical
Species occupancy	Multiple species and communities occupy the ecosystem	Multiple species can occupy same habitat	May or may not be occupied by multiple species	Restricted to only one species
Short definition	Large area in which species exists and interacts	A place within any ecosystem where species is found	Small subset of habitat which is preferred by species	Role of a species in nature
Typical Examples	River, Savanna, Tropical Forest	Stream	Submerged rock in a stream	-

15.Indoor Pollution

Indoor air pollution is dust, dirt, or gases in the air inside buildings such as your home or workplace that could be harmful to breathe in. Poor indoor air quality has been linked to lung diseases like asthma, COPD and lung cancer. It has also been linked to increased risk of heart disease and stroke.



Possible places of indoor pollution with potential pollutants

What are the 4 major indoor air pollutants?

They fall into one of four categories: VOCs (**V**olatile **o**rganic **c**ompounds are emitted as gases from certain solids or liquids), biological pollutants, combustion byproducts, and legacy pollutants. These pollutants can affect the health and comfort of building occupants.

How to cut down your risk?

Fortunately, there are ways you can minimize air pollution in your home, car or at work. Try these simple steps:

- Avoid smoking indoors (but quitting smoking is the best answer for overall health).
- Use craft supplies in well-ventilated areas.
- Make sure your gas stove is well-ventilated.
- Minimize clutter.
- Remove carpeting if possible.
- Use a dehumidifier and/or air conditioner to reduce moisture.
- Keep trash covered to avoid attracting pests.
- Remove shoes at the door.

- Have car emissions tested regularly.
- Minimize air freshener use.
- Test your home for radon.
- Use carbon monoxide detectors.
- Fix water leaks.
- Dust surfaces and vacuum frequently.
- Wash bedding weekly in hot water.
- Make sure exhaust fans are functioning in your bathrooms and kitchen
- Keep a lid on scented candles.

16.Hospital SAFETY and Environment

S – Sense the error
A – Act to prevent it
F – Follow Safety Guidelines
E – Enquire into accidents/Deaths
T – Take appropriate remedial measures
Y – Your responsibility

Following Factors that can help improve patient safety in hospitals

No matter how you define it, patient safety is nothing to take lightly. Each and every person who works in a healthcare facility has a part to play in keeping patients safe. Read up on these factors to discover what you and your healthcare facility can do to help.

1. Use monitoring technology

Healthcare workers care for several patients during their shifts; each one with varying needs and medications. The use of monitoring technology can help nurses and physicians verify that they're always following the right procedure for the right patient and that they're keeping on top of each patient's needs.

2. Make sure patients understand their treatment

You may think all patient safety responsibilities fall to the healthcare provider, but making sure that patients are informed about their own care is also vital to preventing errors. This doesn't mean patients need to have the same knowledge of their situation as a healthcare professional—but a high-level understanding of the treatment and the risks they face can go a long way. Simply asking patients to recall and restate what they've been told is a strategy with enough evidence to support its regular use in patient safety initiatives.

Patients and their family members are naturally focused on maintaining their own health. By verifying that they understand their treatment plan, medication and medical procedures, you're giving them the tools they need to notice and prevent errors in their own care.

3. Verify all medical procedures

We've all heard the horror stories about a patient having their right knee replaced when it was actually the left knee that was scheduled for operation. Building regular verification processes into hospital procedures helps prevent errors like these. The most famous example of this type of verification is the Universal Protocol, which helps prevent surgical errors by allowing any member of a surgical team to call a "time out" to verify the details of the surgery.

4. Follow proper handwashing procedures

Simple actions can have a big impact on patient safety. The Centers for Disease Control and Prevention (CDC) cites handwashing as one of the best ways to stop the spread of germs and prevent infections in hospitals.

The CDC's Clean Hands Count campaign recommends that healthcare providers use alcohol-based sanitizer regularly, as well as have open conversations with patients about the importance of hand hygiene. This allows patients to take control of their own health by asking healthcare providers if they've washed their hands when they enter the room.

5. Promote a team atmosphere

Each and every member of a hospital's staff plays a part in maintaining patient safety. Emphasizing teamwork and being honest with staff about their role in patient safety can make a big difference in your hospital's culture and attitude toward preventing errors. This highlights that the hospitals to share patient feedback and patient safety trends in their hospital, as well as forming interdisciplinary risk assessment teams to help address patient safety issues and promote teamwork among hospital staff.

17. Industrial safety and hazard management

Main Types of Industrial Safety Hazards

Several types of industrial safety hazards can threaten the physical wellbeing of your workers. Five types of risks in the workplace—physical, chemical, ergonomic, biological and safety—could cost your workers their health while decreasing the uptime and productivity of your facility. Make safety a priority at your company instead of an afterthought to see the most significant benefits to your business' employees and its bottom line.

A. Physical

Physical hazards occur when the working area poses a danger to those just standing in it. For instance, noise, radiation exposure and temperature extremes can affect anyone in an industrial area, regardless of if they are working with equipment or not. Because these issues exist in the physical space, they fall under that classification of hazards.

Noise

Noise exposure can cause permanent damage to the ear, leading to life-long hearing loss. Experiencing loud noises for a short while can cause ringing in the ears—tinnitus—that can persist for hours after the exposure. But long-lasting damage can also happen, especially with loud noises or long-term exposure. The damaging effects of noise can occur with long-term exposure to sounds. For example, after eight hours of exposure to noise at only 85 decibels, the level of a lawnmower or vacuum cleaner, can cause damage. Bulldozers and ambulance sirens are louder and can harm hearing after only an hour.

Radiation

Radiation exposure does not only include workers being near uranium or other nuclear powering substances. Microwave and radiofrequency radiation from exposure to magnetic and electrical fields can also harm the body, usually without overt symptoms in the short term. Several studies link these types of fields to cancer. Adhering to the OSHA standards of limiting exposure to no more than 10 milliwatts per square centimeter can protect your company's employees.

Temperature Extremes

Working in unheated, non-air-conditioned facilities or outdoors can expose employees to temperature extremes. Heat and cold both can play havoc on the body's thermoregulation systems.

Heat stress occurs from exposure to high temperatures, which may happen even on chilly days if working near furnaces or other heat-releasing devices. Illnesses that may arise from exposure to excessive heat include rash, cramps, fainting, rhabdomyolysis, exhaustion and **heatstroke.**

The mildest of these conditions, rash and cramps, indicate the sufferer could progressively get worse if he or she stays in the heat. Rhabdomyolysis happens when muscles break down when doing strenuous work in the heat. This condition may lead to electrolyte imbalances and kidney damage. Some people do not experience any symptoms of rhabdomyolysis.

Cold weather can also be dangerous to workers. If your facility has a walk-in freezer or cooler, your company needs a plan for preventing cold-stress illnesses, even in the summer. Cold-related illnesses and injuries include **hypothermia,** frostbite and trench foot.

B. Chemical

Chemical dangers reflect exposure to hazardous chemicals. Anything that could cause illness from inhalation or touch falls under this category. Even innocuous substances in large amounts could lead to physical harm. Providing PPE—personal protection equipment—to your workers who must handle or be around chemicals can protect them from unnecessary exposure.

C. Ergonomic

The human body requires regular exercise and movement. However, doing the same set of actions too often, especially without proper form or support, can lead to overuse injuries. Even long periods of incorrectly standing or sitting can lead to pain. Providing workers with supportive equipment such as back braces and training them in proper movements can help reduce the incidences of musculoskeletal disorders.

D. Biological

Biological hazards come from living organisms, whether those are as tiny as viruses or as large as animals. Illnesses from bacteria, fungi, viruses (e.g, COVID-19), exposure to plants, or animal bites or scratches pose genuine risks for workers. In some cases, these illnesses may be contagious and spread throughout the workforce. Think about how many of your company's workers call in sick during flu season to get an idea of the danger biological hazards have on productivity.

E. Safety

Safety hazards appear while an employee works. Examples of safety hazards include shocks from faulty electrical equipment, slips on floors, falls from working at heights, head injuries from falling debris and cuts or worse from improperly used machinery. To ensure the most effective protection for the workers, follow all job safety guidelines as a baseline. With these requirements in place, the company's workers will be safer and have fewer accidents while your business will avoid fines for failing to adhere to the guidelines.

18.How to Protect the Workers

Once you understand the types of risks your facility's workers face, your company can begin to create means of protecting them in all situations. Ensuring environmental and personal safety will go a long way toward preventing accidents on the floor.

A. **Require Personal Protection Equipment**
Safety equipment separates workers' bodies from hazards in the workplace, which reduces the chances of injuries. Personal protection equipment, PPE, includes hardhats, goggles, hearing protection, gloves and respirators. By requiring appropriate PPE, your company can prevent injuries from exposure to chemicals, biological agents, noise, radiation and accidents. A company should supply safety equipment for their workers. Regularly check the integrity of the PPE they provide and replace any that shows signs of wear or damage. A broken hardhat will not protect a worker from a head injury the same as an intact one will.

B. **Train Employees**
Employees cannot practice safety measures they do not know about. Establish a culture of prioritizing safety that includes regular training and safety updates to keep all employees on the same page. Frequently scheduled training sessions remind workers of the safety steps they must take to avoid accidents and other hazards. When employees have regular reminders, they will likely not become complacent or lax in their work, which can lead to forgetting vital steps in preventing hazardous situations. During the training sessions, ask workers what they need from your business to feel safer on the job. They have first-hand experience and know best the hazards they face.

C. **Prevent Falls**
Protecting workers from falls requires using the proper guardrails and barriers to keep them out of dangerous areas. Keep floors clean and dry to prevent slips and encourage appropriate non-skid footwear for all employees. Authority takes falls very seriously, which is why the organization has such strict requirements for employers to prevent these accidents.

D. **Enforce Safety**
Setting out safety measures only covers part of the equation of protecting the company's workers. Your company's industrial safety system must include a means of enforcing the safety guidelines you establish. Take charge in preventing accidents by having regular checks of the floor and watching whether employees are using proper safety procedures.

E. **Maintain Electrical Equipment**

Poorly maintained electrical equipment poses a serious hazard to operators. Regularly servicing electrical equipment will prevent accidents caused by worn-out wires and other mechanical failings that can cause electrocutions. Instead of waiting for a problem to occur, conduct preventative maintenance and schedule repairs as soon as something minor happens.

19.Environmental Pollution

The major forms of pollution are listed below along with the particular contaminants relevant to each of them:

a. **Air pollution:** the release of chemicals and particulates into the atmosphere. Common gaseous pollutants include carbon monoxide, sulfur dioxide, chlorofluorocarbons (CFCs) and nitrogen oxides produced by industry and motor vehicles. Photochemical ozone and smog are created as nitrogen oxides and hydrocarbons react to sunlight. Particulate matter, or fine dust is characterized by their micrometre size PM10 to PM2.5.
b. **Electromagnetic pollution:** the overabundance of electromagnetic radiation in their non-ionizing form, such as radio and television transmissions, Wi-fi etc. Although there is no demonstrable effect on humans there can be interference with radio-astronomy and effects on safety systems of aircraft and cars.
c. **Light pollution:** includes light trespass, over-illumination and astronomical interference.
d. **Littering:** the criminal throwing of inappropriate man-made objects, unremoved, onto public and private properties.
e. **Noise pollution:** which encompasses roadway noise, aircraft noise, industrial noise as well as high-intensity sonar.
f. **Plastic pollution:** involves the accumulation of plastic products and microplastics in the environment that adversely affects wildlife, wildlife habitat, or humans.
g. **Soil contamination** occurs when chemicals are released by spill or underground leakage. Among the most significant soil contaminants are hydrocarbons, heavy metals, herbicides, pesticides and chlorinated hydrocarbons.
h. **Radioactive contamination,** resulting from 20th century activities in atomic physics, such as nuclear power generation and nuclear weapons research, manufacture and deployment. (See alpha emitters and actinides in the environment.)
i. **Thermal pollution,** is a temperature change in natural water bodies caused by human influence, such as use of water as coolant in a power plant.

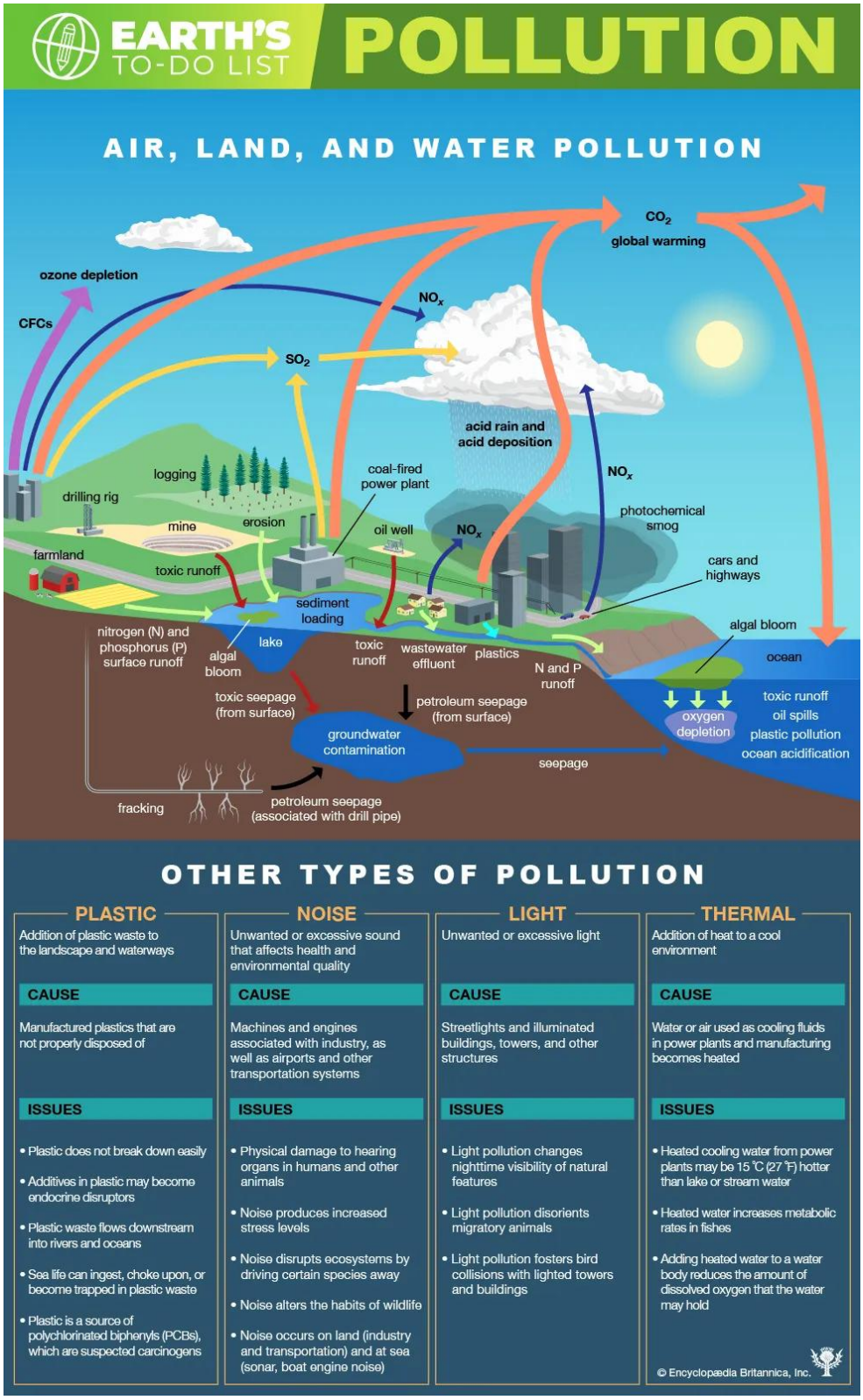
j. **Visual pollution,** which can refer to the presence of overhead power lines, motorway billboards, scarred landforms (as from strip mining), open storage of trash, municipal solid waste or space debris.
k. **Water pollution,** caused by the discharge of industrial wastewater from commercial and industrial waste (intentionally or through spills) into surface waters; discharges of untreated sewage and chemical contaminants, such as chlorine, from treated sewage; and releases of waste and contaminants into surface runoff flowing to surface waters (including urban runoff and agricultural runoff, which may contain chemical fertilizers and pesticides, as well as human feces from open defecation).

20. Effects on Environment:

Pollution has been found to be present widely in the environment. There are a number of effects of this:

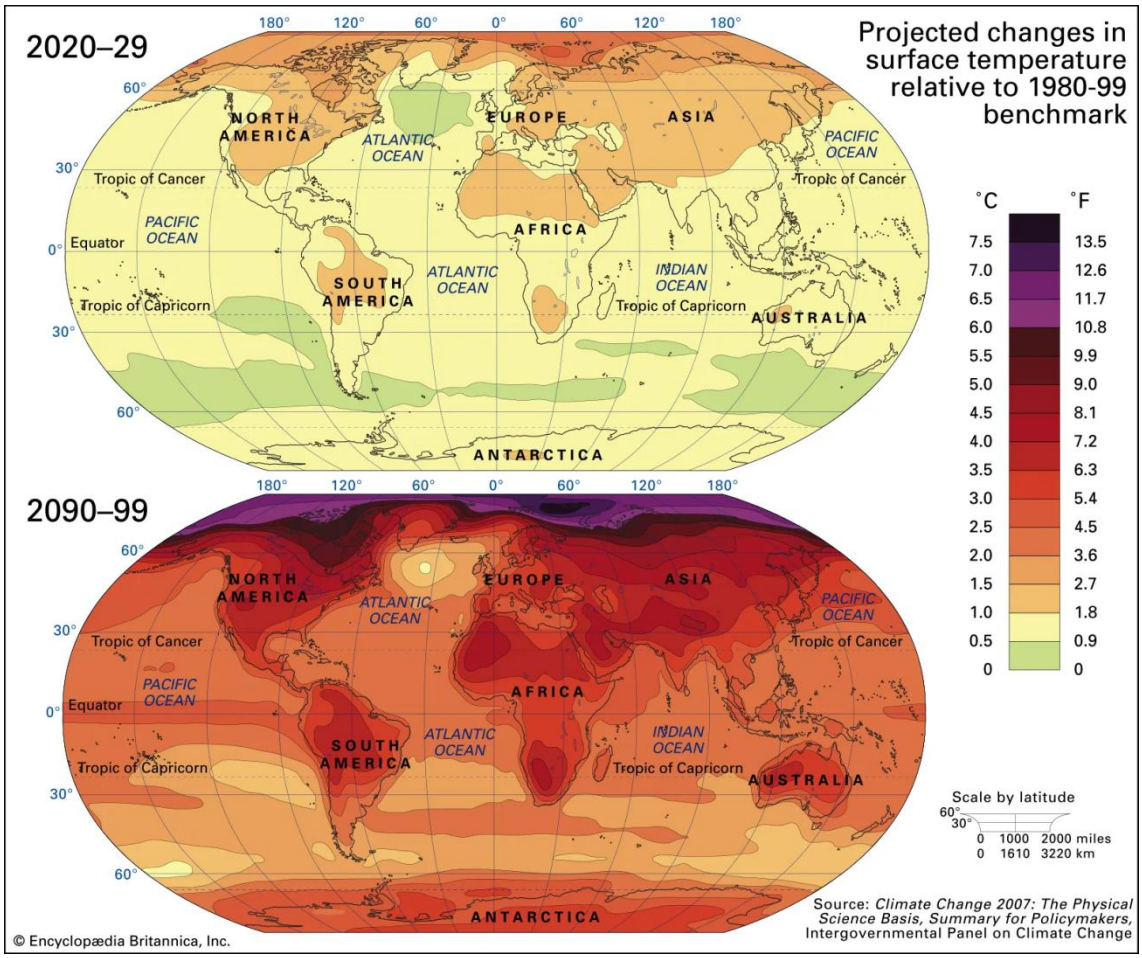
- Biomagnification describes situations where toxins (such as heavy metals) may pass through trophic levels, becoming exponentially more concentrated in the process.
- Global carbon dioxide emissions by jurisdiction (as of 2015)
- Carbon dioxide emissions cause ocean acidification, the ongoing decrease in the pH of the Earth's oceans as CO2 becomes dissolved.
- The emission of greenhouse gases leads to global warming which affects ecosystems in many ways.
- Invasive species can outcompete native species and reduce biodiversity. Invasive plants can contribute debris and biomolecules (allelopathy) that can alter soil and chemical compositions of an environment, often reducing native species competitiveness.
- Nitrogen oxides (NO_2 or NO_x) are removed from the air by rain and fertilise land which can change the species composition of ecosystems.
- Smog and haze can reduce the amount of sunlight received by plants to carry out photosynthesis and leads to the production of tropospheric ozone which damages plants.
- Soil can become infertile and unsuitable for plants. This will affect other organisms in the food web.
- Sulfur dioxide (SO_2) and nitrogen oxides can cause acid rain which lowers the pH value of soil.
- Organic pollution of watercourses can deplete oxygen levels and reduce species diversity.

21.Effects of Pollution in Diagram



22. Global Warming

Global warming is the long-term heating of Earth's surface observed since the pre-industrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere.



23. Effects of Global Warming

a. Destruction of Ecosystems

Global warming destabilizes natural ecosystems by altering temperature and rainfall, leading to habitat loss and species imbalance.

📍 *Example from Bangladesh:* The **Sundarbans mangrove forest** is increasingly threatened by rising salinity and erratic weather, which harm its unique flora and fauna—especially the Royal Bengal Tiger's habitat.

b. Loss of Biodiversity

As climate shifts occur, many species lose their habitats or fail to adapt, leading to extinction or migration.

📍 *Example from Bangladesh:* Indigenous freshwater fish species like **Pabda and Baila** are declining due to rising water temperatures and pollution in rivers and wetlands across northern and central Bangladesh.

c. Conflicts and War

Climate-induced scarcity of water, land, or food can worsen tensions and even trigger social unrest.

📍 *Example from Bangladesh:* In **coastal areas like Satkhira**, reduced agricultural productivity due to salinity intrusion has caused local land disputes and migration to urban slums, straining urban infrastructure and employment.

d. Economic Consequences

Climate change leads to rising costs in agriculture, healthcare, disaster management, and infrastructure repair.

📍 *Example from Bangladesh:* The **2017 floods in northern Bangladesh** caused significant crop damage (rice and jute), resulting in massive economic losses for poor farmers and requiring emergency government aid.

e. Disease

Warmer and wetter climates enable disease-carrying insects and microbes to thrive, increasing health risks.

📍 *Example from Bangladesh:* The **spread of dengue fever** in Dhaka has intensified over recent years, with warmer winters and increased rainfall extending mosquito breeding seasons.

f. Drought

Prolonged dry periods reduce water availability and agricultural productivity, threatening food and water security.

📍 *Example from Bangladesh:* The **Barind Tract** in Rajshahi and Chapainawabganj frequently experiences droughts, affecting boro rice cultivation due to water scarcity and over-dependence on groundwater irrigation.

g. Storms and Floods

Global warming increases the intensity and frequency of extreme weather events, such as cyclones and heavy rainfall.

📍 *Example from Bangladesh:* **Cyclone Amphan (2020)** caused widespread flooding and destruction in the Khulna and Satkhira districts, displacing thousands and salinizing large tracts of farmland.

h. Heat Waves

Higher temperatures result in severe heat waves, leading to health hazards, reduced productivity, and energy stress.

📍 *Example from Bangladesh:* The **record-breaking heatwaves of April 2023** led to school closures in several districts and increased hospital admissions due to heat-related illnesses.

i. Shrinking Glaciers

Melting glaciers threaten long-term freshwater supplies and river flow patterns downstream.

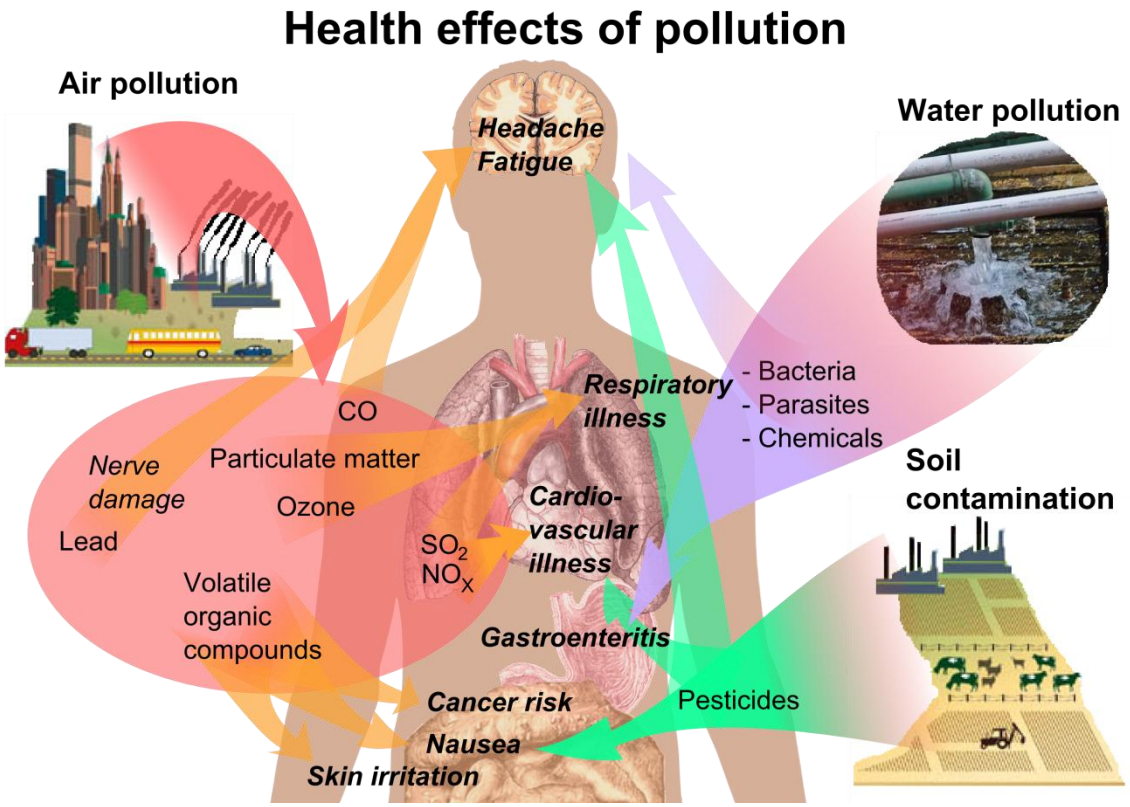
📍 *Example from Bangladesh:* Though Bangladesh has no glaciers, it **depends on rivers** originating in the **Himalayas**, such as the Ganges and Brahmaputra. Himalayan glacier retreat alters water flow, impacting agriculture and fisheries in Bangladesh's floodplains.

j. Rising Sea Level

Sea-level rise causes coastal erosion, salinity intrusion, and displacement of people.

📍 *Example from Bangladesh:* In **Bhola and Bagerhat**, rising seas have submerged farmlands, forced people to migrate inland, and increased the number of "climate refugees" seeking work in cities like Dhaka.

24. General Human Health Effects of Pollution



25. Reproductive Health Hazards in Environmental Science

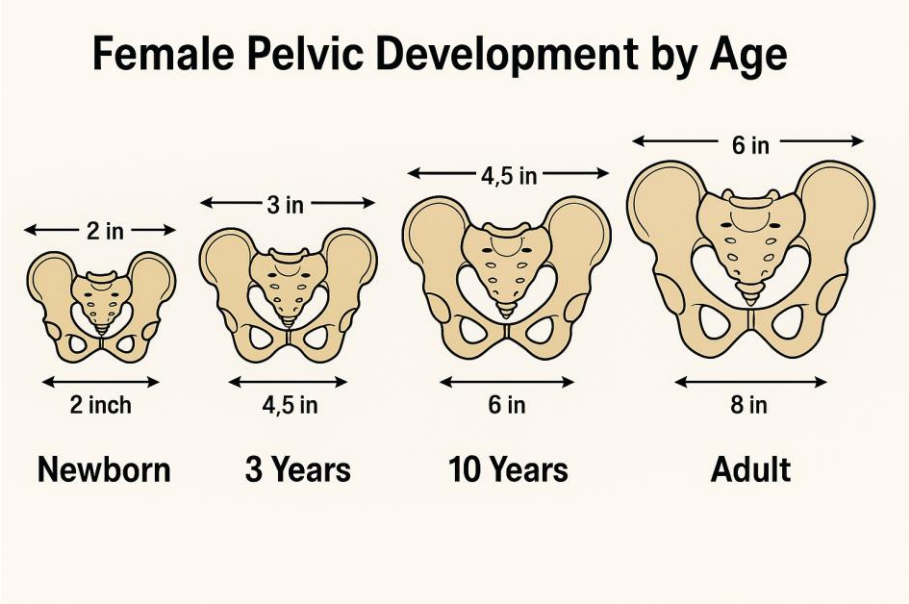
The **Environmental Science course**, though primarily focused on ecological and natural systems, inherently intersects with **human health, social behavior**, and **population dynamics**—all of which are influenced by hormonal changes and marital status. In Bangladesh, where **demographic transitions, reproductive health**, and **gendered environmental roles** are deeply embedded in societal structures, understanding hormone-driven physiological changes (such as puberty, fertility, and menopause) provides essential context for discussing **population growth, maternal health, and environmental stress**. For instance, early puberty in girls, often linked to poor nutrition or environmental pollutants (like endocrine disruptors), may accelerate early marriage and childbirth—factors that directly influence household size, land use, and food demand.

According to UNICEF and BBS, **over 50% of girls in Bangladesh are married before the age of 18**, especially in rural and low-income areas. Early marriage often results in early pregnancy—before full pelvic or psychological development—leading to:

- High rates of **maternal mortality**
- **Low birth weight** and malnourished infants
- Long-term **gynecological complications** such as **obstetric fistula**

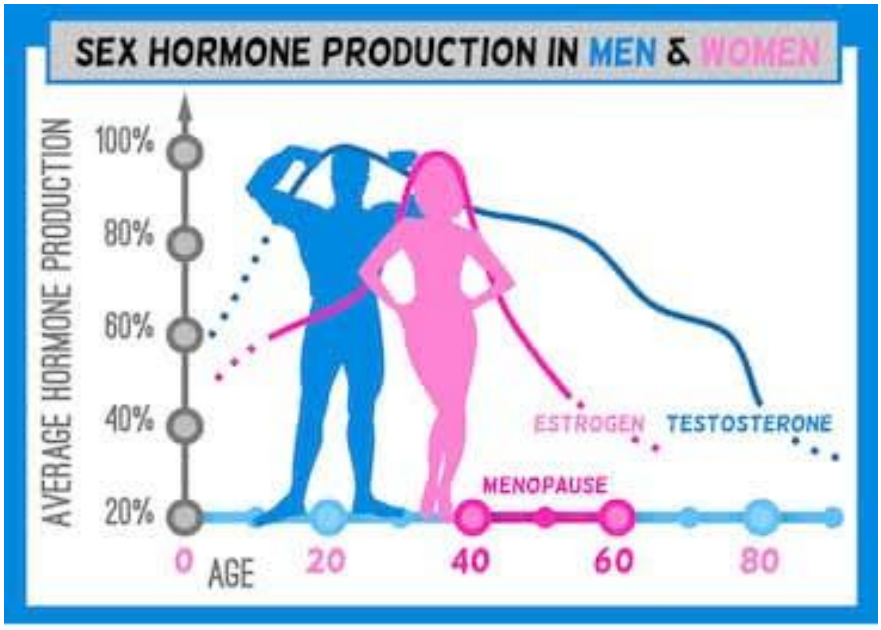
In **climate-vulnerable areas** (e.g., coastal regions), **poverty, dowry pressure**, and **displacement** increase the incidence of early marriage. The **female pelvic bone** (particularly the **pelvic girdle**) does **not fully mature until approximately 22 years of age**. Before that, a girl's **skeletal system—especially the pelvic inlet—is underdeveloped** for safe childbirth.

Biological Area	Description
Pelvic Maturity	The female pelvis grows and reshapes significantly during adolescence. Full development usually completes by age 20**–22**.
Hormonal Influence	Estrogen plays a major role in widening the pelvis during puberty, supporting future childbirth.
Risks of Early Pregnancy	Immature pelvic bones increase risk of **obstructed labor**, **fistula**, and **maternal mortality**.
WHO Recommendation	WHO strongly discourages pregnancy before age **18**, both for physical and psychological readiness.



Biologically, this course also explores how **girls' bodies are not physically prepared for childbirth during adolescence** due to incomplete **pelvic bone growth**. The underdeveloped pelvic structure, typically not mature until age 20 or later, significantly increases the risk of **obstructed labor, maternal mortality**, and **long-term gynecological damage** when girls become pregnant prematurely. By integrating these aspects, Environmental Science education empowers students to connect **human reproductive health** with environmental pressures and societal outcomes—revealing that sustainable development requires not only ecological balance but also **gender-sensitive, age-appropriate, and inclusive population policies**, especially in countries like Bangladesh.

Furthermore, marital status in Bangladesh is often tied to **social expectations of reproduction**, especially in women, where early marriage remains common despite national campaigns. This cultural norm, paired with limited reproductive education, leads to early pregnancies before full hormonal and skeletal maturity—resulting in **health risks and strain on public health infrastructure**. These dynamics affect population distribution, urban migration, and resource use. By including such gendered and biological perspectives in Environmental Science, students are better equipped to analyze **how human behavior and reproductive health link back to environmental sustainability**, social justice, and climate vulnerability.



Parameter	Women (Estrogen)	Men (Testosterone)
Rise in puberty	Rapid (age 10–12)	Steady (age 12–14)
Peak production	20–30 years	25–40 years
Decline begins	Around 35 years	After 45 years
Sharp hormonal drop	Yes (due to menopause)	No (gradual decline)
Reproductive implication	Sudden end of fertility	Slower decline in fertility

Environmental legacy refers to the **long-term, often invisible, impacts of past industrial, agricultural, and urban practices** that leave behind pollutants in air, soil, water, and food systems. This has critical implications:

- 🌸 **Irregular Menstrual Cycles**: Linked with BPA and phthalate exposure.
- 🧬 **Ovarian dysfunction and PCOS-like symptoms**: Triggered by hormonal imbalance caused by persistent pollutants.
- 👶 **Infertility and Miscarriage Risk**: Due to ovulatory disorders and poor implantation environments.
- 👧 **Early Puberty (Precocious Puberty)**: Environmental estrogens accelerate maturation, particularly in urban girls.
- 🩸 **Increased Risk of Endometriosis and Cancer**: Especially with long-term exposure to dioxins and PCBs.

A **higher than normal vaginal pH** (3.4-3.8) creates a less protective environment, especially during menstruation, pregnancy, or intercourse, increasing susceptibility to:

- 🦠 **Infections (UTIs, BV, Candida)**
- 🤰 **Pregnancy complications**
- 🧪 **Sperm mortality and reduced fertility**

It is evident that how **environmental pollution**, once considered an industrial issue, has **profound biological and reproductive consequences**—especially for vulnerable populations like women and adolescent girls. This supports a **gender-inclusive approach** to sustainability, public health, and policy development.

Chemical	Common Use	Risk Area
BPA *(Bisphenol A)*	Found in plastic bottles, food cans	Can mimic estrogen and affect puberty and fertility
Phthalates	Used in plastics, cosmetics, toys	Disrupt testosterone levels and male reproductive development
DDT Dichloro-Diphenyl-Trichloroethane	Pesticide (banned in many countries)	Long-lasting reproductive and neurological effects for both male and female
PCBs *Polychlorinated Biphenyls*	Industrial chemicals in old electronics and paints	Affect thyroid hormones and brain development
Atrazine	Common agricultural herbicide	Associated with reproductive deformities in amphibians and humans

This **course** is intricately linked with **early marriage, pelvic bone growth, child marriage**, and the **environmental factors** that influence these issues, especially within the context of **Bangladesh's socio-ecological systems**. In regions affected by **poverty, climate stress**, and **resource scarcity**—such as flood-prone coastal belts or drought-prone northern districts—families often view early marriage as a **coping strategy** for economic survival or protection from insecurity. This environmental vulnerability, coupled with social norms, contributes to **high rates of child marriage**. The course enables students to explore how **environmental degradation and climate disasters** (e.g., cyclones, salinity, river erosion) can indirectly drive early marriage by pushing families into migration, poverty, and marginalization.